

Predicting Cardiac Diseases in Children

Introduction do Data Science Report

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Master in Artificial Intelligence

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12/2025

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1 Raw data

1.1 Taxonomy of Data

The dataset consists of 21 variables, including demographic, clinical, and diagnostic data. Categorizing these variables is essential as specific statistical methods and distance measures for cluster analysis depend on the variable type.

1.1.1 Ratio-Scaled Variables

These continuous variables represent physical measurements with a true zero:

- **Physical Metrics:** `Peso` (Weight), `Altura` (Height), and `IMC` (Body Mass Index).
- **Vital Signs:** `IDADE` (Age), `PA SISTOLICA` (Systolic Pressure), `PA DIASTOLICA` (Diastolic Pressure), and `FC` (Cardiac Frequency).

1.1.2 Binary Variables

- **Target Variable:** `NORMAL X ANORMAL`, representing the presence or absence of a cardiac pathology.

1.1.3 Nominal Variables

These categorical variables do not have an intrinsic ordering:

- **Demographics/Admin:** `SEX0` (Gender) and `Convenio` (Insurance).
- **Clinical Findings:** `SOPRO` (Murmur type), `B2` (Second heart sound characteristics), and `PULSOS` (Pulse palpability).
- **Medical History:** `MOTIV01/2` (Referral reasons) and `HDA1/2` (History of disease).

1.1.4 Ordinal Variables

- **PPA (Pediatric Pressure Assessment):** This variable classifies blood pressure into ordered clinical stages (Normal, Pre-Hypertension, Hypertension Level 1, etc.).

1.2 Preliminary Analysis

The preliminary analysis was performed on the raw dataset to "uncover" structural inconsistencies, outliers, and noise before any cleaning was applied. This stage is critical for justifying the subsequent preprocessing steps.

1.2.1 Statistical Summary and Spread Measures

We calculated the central tendency and dispersion for the numerical clinical variables. As observed in Table 1, the presence of extreme ranges suggests significant data entry errors.

Table 1: Summary of Spread Measures for Raw Numerical Variables

Variable	Mean	Std Dev	Min	Max	Range	IQR
Peso (kg)	21.16	18.06	-40.00	157.00	197.00	24.40
Altura (cm)	83.87	56.58	0.00	198.00	198.00	94.00
IDADE (years)	6.01	7.83	-113.18	71.81	184.99	7.79
PA Sistólica	101.31	15.51	10.00	990.00	980.00	20.00
FC (bpm)	93.80	100.43	8.00	9288.00	9280.00	20.00

1.2.2 Numerical Distributions and Outliers

The visualization of numerical data through histograms and boxplots revealed several key inconsistencies:

- **Density and Skewness:** Figure 4 shows that **Peso** and **IDADE** are heavily right-skewed. The density plots also highlight a spike at 0.0 for **Altura**, indicating that missing values were likely recorded as zeros.
- **Physiological Outliers:** As seen in Figure 2, the Heart Rate (**FC**) and Systolic Pressure (**PA**) contain extreme outliers (e.g., 9288 bpm) which are physiologically impossible and represent clerical errors.
- **Negative Values:** The boxplots confirm the existence of negative values for age and weight, which are logically invalid.

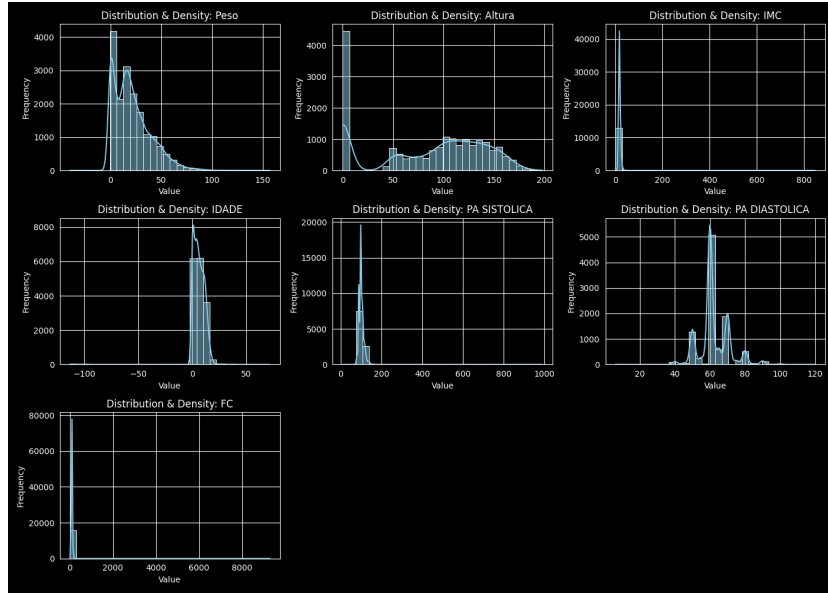


Figure 1: Histograms and Density plots of the raw numerical features.

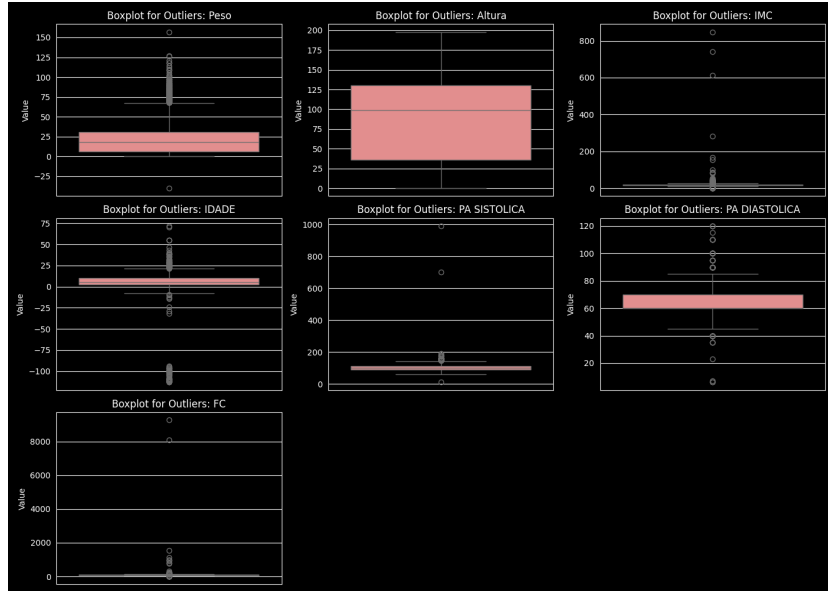


Figure 2: Boxplot analysis used to uncover extreme outliers and invalid ranges.

1.2.3 Categorical Label Noise

An analysis of categorical frequencies (Figure 31) uncovered "label noise" that artificially increases the dimensionality of the data.

- **Synonyms and Inconsistent Casing:** Several variables exhibit label duplication where identical categories are treated as distinct entities. For instance, in the `SEX0` variable, "M", "Masculino", and "masculino" all represent the same class. This issue is recurrent across the dataset, affecting other features such as `NORMAL X ANORMAL` (e.g., "Anormal" vs. "anormal") and `SOPRO` (e.g., "Sistólico" vs. "sistólico").
- **Formula Errors:** The `PPA` variable contains the string "#VALUE!", suggesting a broken Excel formula in the source file.

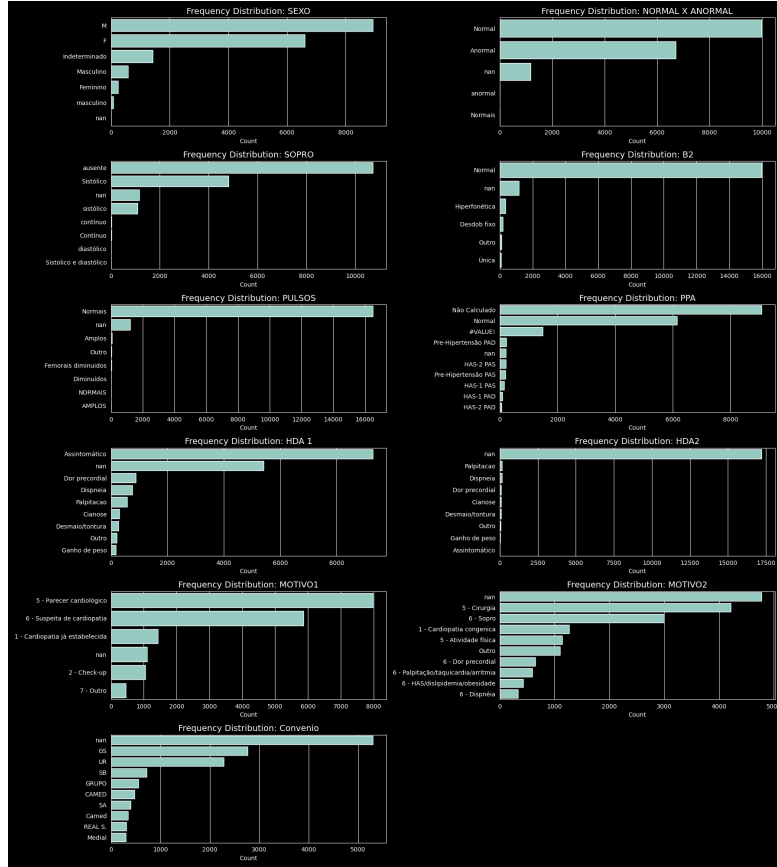


Figure 3: Frequency distribution of nominal and binary variables, highlighting label duplication.

1.3 Main changes needed to be performed

To prepare the raw data for modeling, a rigorous preprocessing pipeline was established. The following changes were implemented to address the inconsistencies and noise "uncovered" during the preliminary analysis:

1. Feature Selection and Dimensionality Reduction:

- **Redundant Variables:** The PPA (Pediatric Pressure Assessment) column was removed as it is a derived categorical representation of blood pressure readings already present in the dataset.
- **Irrelevant Metadata:** The patient ID, visit date (*Atendimento*), and birth date (*DN*) were excluded as they possess no predictive value for cardiac pathology.
- **High Missingness:** The variable *HDA2* was dropped entirely due to a missing value rate of 96%, which renders it statistically insignificant.

2. Clinical and Physiological Filtering:

To ensure the model learns from valid biological data, records were filtered based on pediatric physiological bounds:

- **Age & Growth:** Data was restricted to children aged between 2 and 19 years. Records with impossible weight ($\leq 2\text{kg}$ or $> 100\text{kg}$), height ($< 50\text{cm}$ or $> 190\text{cm}$), or IMC (< 10 or > 60) were removed.
- **Vital Signs:** Heart rate (FC) was limited to a physiological range of 40–220 bpm.
- **Blood Pressure Integrity:** Records were required to have both systolic and diastolic readings, following the medical constraint that systolic pressure must be greater than or equal to diastolic pressure, within ranges of 50–200 mmHg and 20–150 mmHg respectively.

3. Categorical Standardization:

- **Gender Consolidation:** Inconsistent labels for gender (e.g., "Masculino", "masculino") were mapped to a standard binary format ("M" or "F").
- **Text Normalization:** All categorical strings were stripped of leading/trailing whitespace and converted to a uniform case to merge identical categories.
- **Handling Rare Categories:** To prevent overfitting, rare categories in *SOPRO* (< 20 occurrences) and *Convenio* (< 80 occurrences) were grouped into a generic "OUTROS" (Others) category.

4. Data Integrity and Row-wise Cleaning:

- **Duplicates:** Exact duplicate rows were removed to ensure each clinical visit is unique and to prevent data leakage.

- **Missing Values:** Any record missing more than 50% of its feature values was discarded. Additionally, rows missing the target variable (`NORMAL X ANORMAL`) were removed, as they cannot be used for supervised learning.

2 Preprocessed data

Following the execution of the cleaning pipeline, the dataset was transformed from a noisy clinical export into a curated structure suitable for statistical modeling and machine learning.

2.1 Final Dataset Characteristics

The resulting preprocessed dataset contains **7233** observations and **16** variables. This reduction from the original 20 variables reflects the strategic removal of redundant or high-missingness features (`PPA`, `HDA2`, `DN`, `Atendimento`).

2.2 Data Integrity Validation

A secondary validation check was performed to ensure the success of the pre-processing steps:

- **Numerical Consistency:** All vital signs and physical metrics now fall within strict pediatric physiological boundaries. There are no remaining zero or negative values for `IDADE`, `Peso`, or `Altura`.
- **Categorical Uniformity:** The `SEX0` variable is now strictly binary (`M`, `F`). All text-based features are standardized to uppercase with no special characters or trailing spaces.
- **Reduced Sparsity:** By grouping rare categories in `Convenio` and `SOPRO` into an `'OUTROS'` class, the feature space has been consolidated, reducing the risk of model overfitting to low-frequency noise.

2.3 Final Variable Types

The table below summarizes the final state of the features that will be used for the subsequent Bivariate and Multivariate analysis.

Table 2: Final Feature Set after Preprocessing

Feature Name	Measurement Scale	Data Type
IDADE, Peso, Altura, IMC	Ratio-Scaled	Float
PA SISTOLICA, PA DIASTOLICA, FC	Ratio-Scaled	Float
SEXO	Binary	Categorical
NORMAL X ANORMAL	Binary (Target)	Categorical
SOPRO, B2, PULSOS, Convenio	Nominal	Categorical
MOTIVO1, MOTIVO2, HDA1	Nominal	Categorical

3 Basic Description

In this section, we provide a statistical overview of the dataset after cleaning and preprocessing. The analysis focuses on the central tendency, spread, and density of the variables, which are essential for understanding the underlying pediatric population.

3.1 Numerical Variables: Spread and Density

The numerical distributions, visualized through histograms and density plots in Figure 4, reveal several key physiological patterns:

- **Weight (Peso) and BMI (IMC):** Both variables exhibit a significant *right-skewed* distribution. While the majority of patients are concentrated in the 10–40 kg range for weight and 15–20 for IMC, a long right tail represents a smaller, clinically relevant group of heavier or obese adolescents.
- **Height (Altura):** This variable shows a wide distribution (50–190 cm) with a *mild left skew*. This reflects the pediatric growth curve across the 2–19 age range, with a higher density of observations in the adolescent height brackets compared to very young children.
- **Age (Idade):** The distribution is multimodal, reflecting the specific demographic peaks of the clinic’s pediatric population. There is a higher concentration of school-aged children and early adolescents, with fewer patients at the age extremes (2 and 19 years).
- **Blood Pressure (PA Sistólica/Diastólica):** Both variables follow a central, approximately normal distribution. Most values fall within the expected pediatric norms (SBP $\approx$ 90–110; DBP $\approx$ 50–70), confirming that the data cleaning successfully removed physiologically impossible outliers.
- **Heart Rate (FC):** The density plot for heart rate is notably irregular and multimodal, with peaks between 70 and 120 bpm. This lack of a single central peak is expected in pediatric data, as “normal” heart rates vary significantly across different developmental stages (e.g., a normal rate for a toddler is considered tachycardia for a late adolescent).

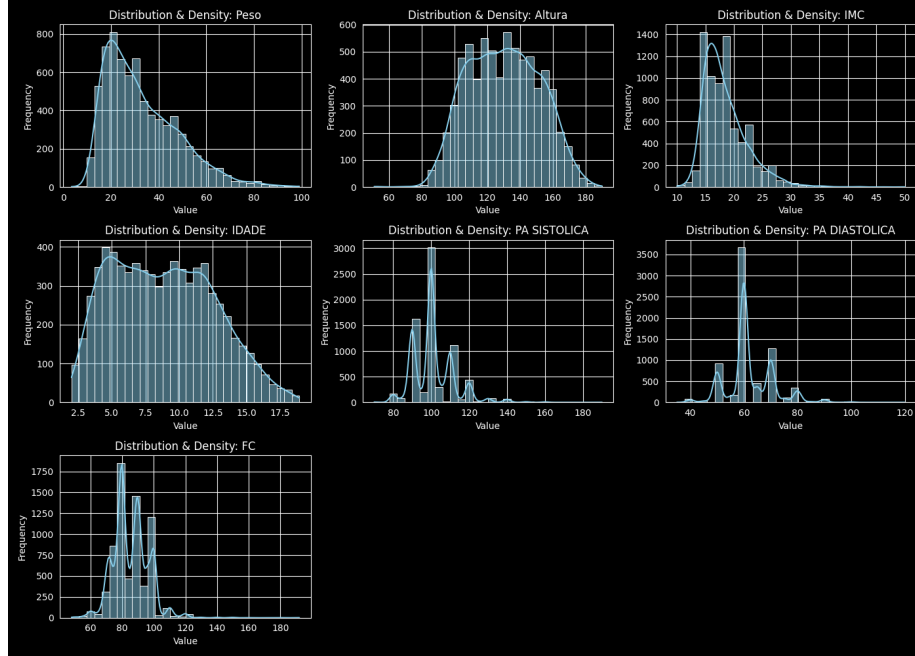


Figure 4: Distributions and density of numerical clinical variables.

3.2 Categorical Variables: Frequency and Imbalance

The categorical features, shown in Figure 5, provide insight into the clinical and demographic makeup of the sample:

- **Gender (SEXO):** The distribution is relatively balanced, with males representing 58.9% and females 41.1% of the records.
- **Clinical Indicators (SOPRO, B2, PULSOS):** These variables exhibit strong class imbalance. Most patients present "Normal" or "Ausente" (absent) results for murmurs, second heart sounds, and pulses. However, the minority classes (e.g., *Sistólico* murmurs or *Amplas* pulses) are retained as they carry high diagnostic value.
- **High-Cardinality Variables (CONVENIO, HDA1, MOTIVO):** These variables originally contained hundreds of unique entries. After preprocessing, dominant categories were identified, while rare categories were grouped into "OUTROS" to avoid excessive sparsity in the predictive models.

3.3 Target Variable and Evaluation Strategy

The target variable **NORMAL_X_ANORMAL** is moderately imbalanced, with the majority of the population classified as "NORMAL."

Because a simple accuracy metric would be deceptively high by favoring the majority class, this study will prioritize **Precision**, **Recall**, **ROC-AUC** and the **F1-score**. These metrics ensure that the model is penalized for failing to detect the minority "ANORMAL" class, which is the primary goal of cardiac screening.

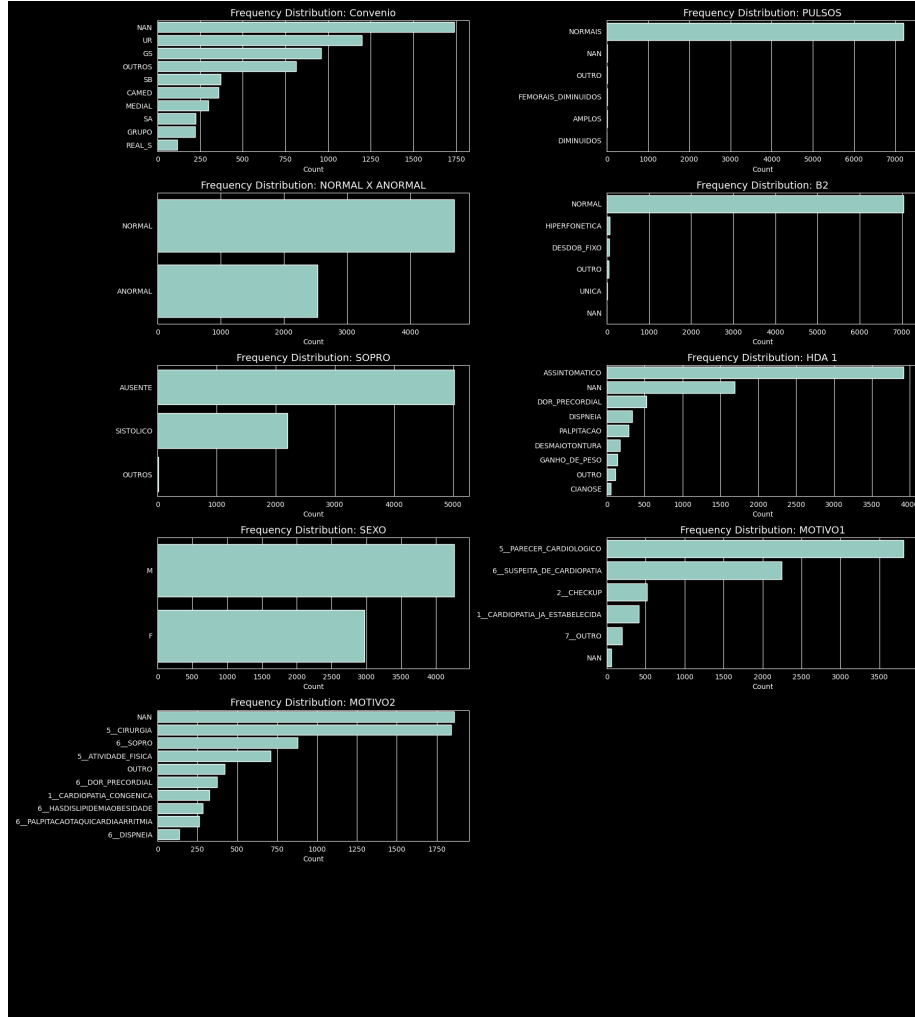


Figure 5: Frequency distributions of categorical features and the target variable.

4 Analysis

4.1 Bivariate analysis

After preprocessing, bivariate analysis was performed to (i) identify **redundant variables** (strong dependencies that suggest overlapping information), and (ii) quantify how strongly each feature is associated with the **target** NORMAL X ANORMAL, guiding an informed pre selection before multivariate methods and predictive modeling. Given that the dataset combines **numerical** variables and **categorical** variables, different bivariate techniques were used depending on the variable type.

4.1.1 Numerical - Numerical relationships (correlation structure)

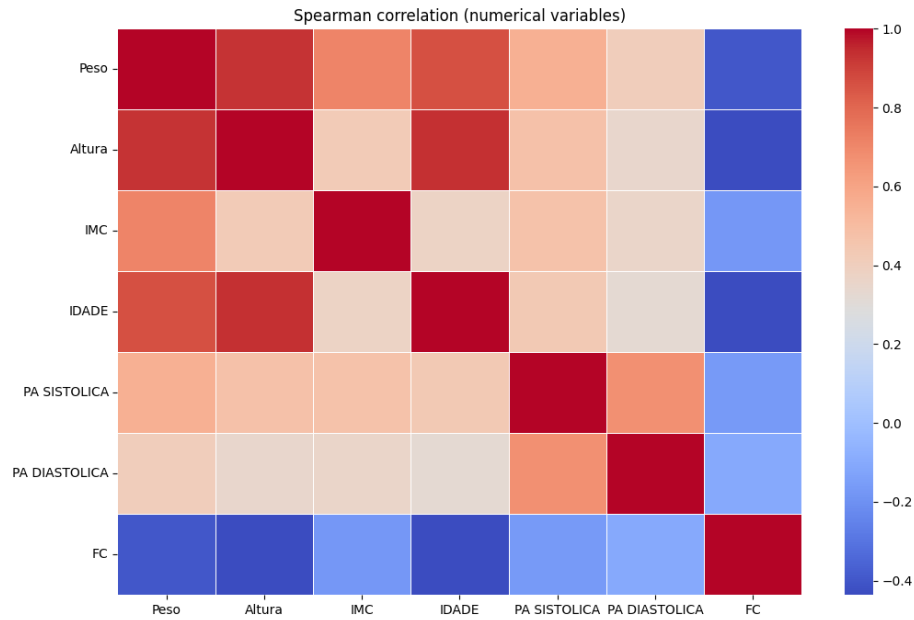


Figure 6: Spearman Correlation Matrix of Numerical Variables

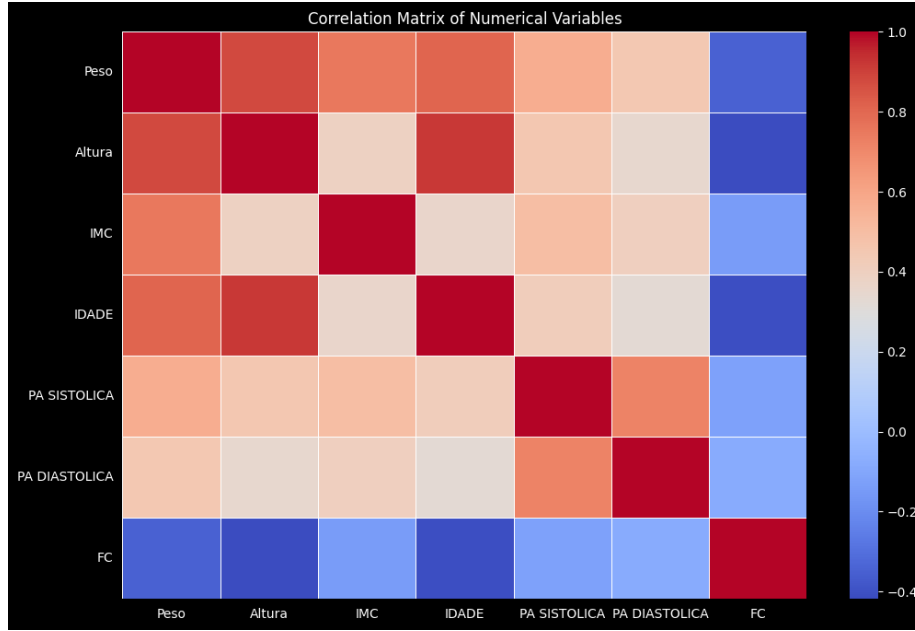


Figure 7: Pearson Correlation Matrix of Numerical Variables

For numerical variables, we computed two correlation matrices (Figures 6 and 7). Because some physiological relationships can be non-linear and numerical variables may still exhibit mild skewness, we report Spearman’s rank correlation as the primary measure and Pearson correlation as a complementary indicator of linear dependence. In our data, both matrices are highly consistent, suggesting that the main dependencies between numerical variables are largely monotonic and approximately linear.

This analysis is particularly relevant in this dataset because some variables are expected to be structurally related:

- **PESO, ALTURA, IMC:** IMC is mathematically derived from weight and height, so moderate-to-strong dependence is expected. Strong correlation patterns indicate potential redundancy, suggesting that downstream feature selection should avoid keeping all three variables without justification.
- **PA SISTOLICA vs. PA DIASTOLICA:** These measures tend to covary physiologically; strong correlation supports that they share cardiovascular signal, though they may still contribute differently to pathology prediction.
- **IDADE with growth/vitals:** Age is expected to relate to height and weight, and may partially explain variation in heart rate and blood pressure in pediatrics, which is useful for interpreting later model behavior.

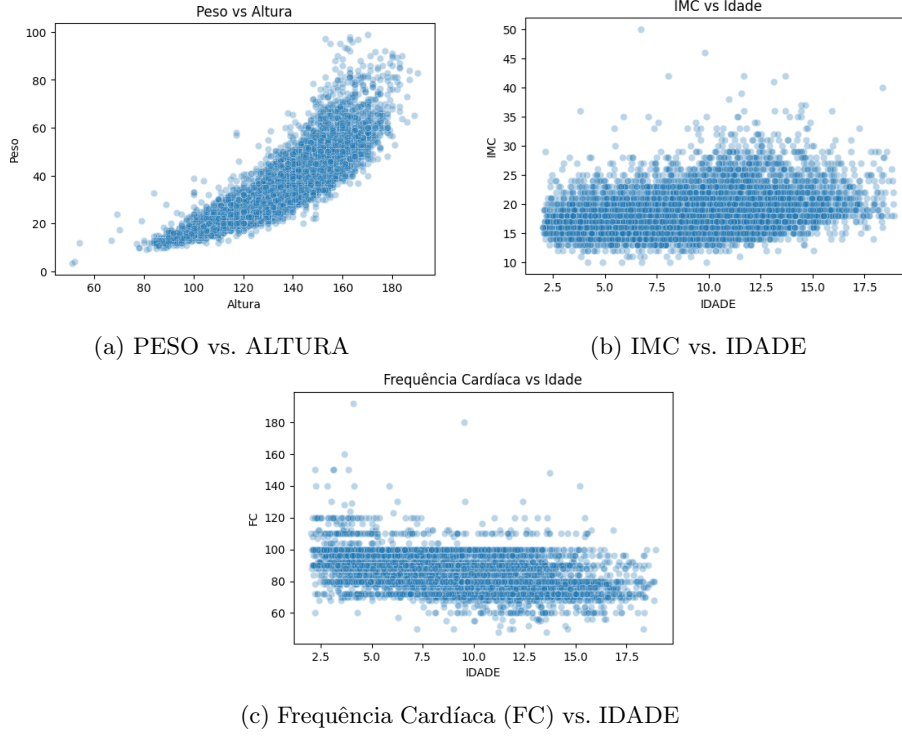


Figure 8: Scatterplots for Key Numeric Relationships

To complement the correlation heatmaps, we inspected scatterplots for a small set of representative and clinically meaningful numerical pairs (Figure 8). The goal was twofold: (i) confirm that the strongest associations observed in the matrices reflect genuine global trends rather than a small number of extreme cases, and (ii) assess the *shape* of the relationships (e.g., linear vs. curved patterns and heteroscedasticity), which is not captured by a single correlation coefficient.

The **PESO vs. ALTURA** plot shows a clear positive association consistent with pediatric growth, with an evident fan-shaped dispersion (variance increasing with height), indicating that the relationship is strong but not perfectly linear. The **IMC vs. IDADE** plot, in contrast, displays a much weaker and more diffuse pattern: IMC values spread across all ages with only a mild trend, suggesting that age alone does not explain body composition and supporting the near-neutral correlations observed in the heatmaps. Finally, **FC vs. IDADE** reveals an overall decreasing tendency of heart rate with age, together with substantial variability and a few extreme values; this confirms the negative association measured in the correlation matrices while highlighting the natural heterogeneity of vital signs across pediatric patients.

4.1.2 Numerical - Target relationships

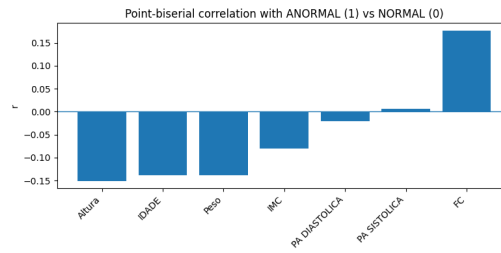
To evaluate how each numerical variable relates to the binary outcome *NORMAL X ANORMAL*, we used two complementary perspectives. First, we summarized group-wise central tendency and dispersion (mean, median, standard deviation) for each variable. Second, we quantified association using the **point-biserial correlation**, which measures linear association between a continuous variable and a binary class label (encoded as 0/1).

		NORMAL X ANORMAL	ANORMAL	NORMAL
IDADE	mean		8.102	9.192
	median		7.580	9.255
	std		3.643	3.748
Peso	mean		30.062	34.486
	median		26.000	31.000
	std		14.627	15.374
Altura	mean		125.764	132.581
	median		123.000	133.000
	std		21.235	21.328
IMC	mean		17.996	18.644
	median		17.000	18.000
	std		3.928	3.779
PA SISTOLICA	mean		101.138	100.982
	median		100.000	100.000
	std		11.574	10.491
PA DIASTOLICA	mean		61.911	62.274
	median		60.000	60.000
	std		8.259	8.731
FC	mean		88.172	84.055
	median		90.000	80.000
	std		11.485	10.642

Figure 9: Group-wise descriptive stats

	Variable	PointBiserial_r	p_value
2	Altura	-0.151034	3.610475e-38
0	IDADE	-0.138871	1.775764e-32
1	Peso	-0.138329	3.103638e-32
3	IMC	-0.080430	7.369949e-12
5	PA DIASTOLICA	-0.020189	8.599121e-02
4	PA SISTOLICA	0.006852	5.601454e-01
6	FC	0.176756	1.019839e-49

(a) Point-Biserial correlation and p-value



(b) Point-Biserial correlation plot

Figure 10: Point-Biserial correlation ANORMAL (1) vs NORMAL (0)

Overall, numerical variables show weak individual association with the diagno-

sis (all $|r| < 0.20$), indicating that no single vital sign or growth measure is a strong stand-alone discriminator. Nonetheless, consistent directional trends emerge. The *ANORMAL* group tends to be **younger** and **smaller** on average (lower IDADE, PESO, and ALTURA), while FC (heart rate) is **higher** in *ANORMAL*. This is reflected in the point-biserial results: FC has the largest positive association with the target ($r \approx 0.18$), whereas ALTURA, IDADE, and PESO show small negative associations ($r \approx -0.14$ to -0.15). In contrast, PA SISTOLICA and PA DIASTOLICA are nearly indistinguishable across classes (near-zero correlation and very similar medians), suggesting limited diagnostic contribution from blood pressure in isolation.

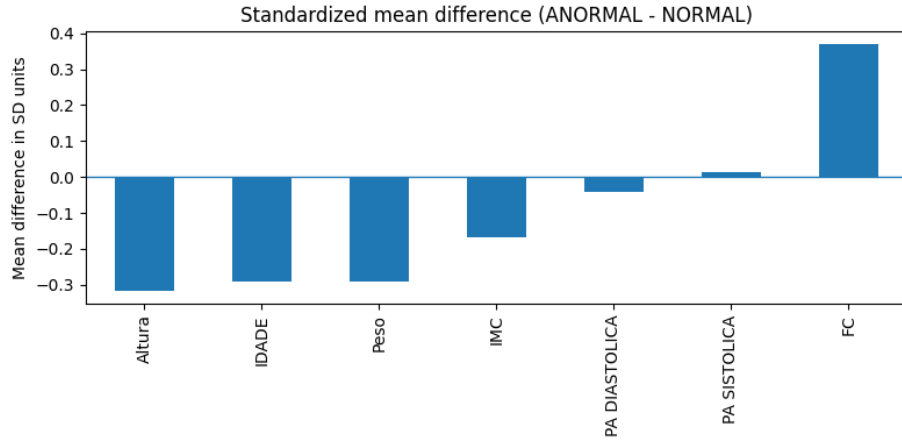


Figure 11: Standardized Mean Difference (ANORMAL - NORMAL) across numerical variables

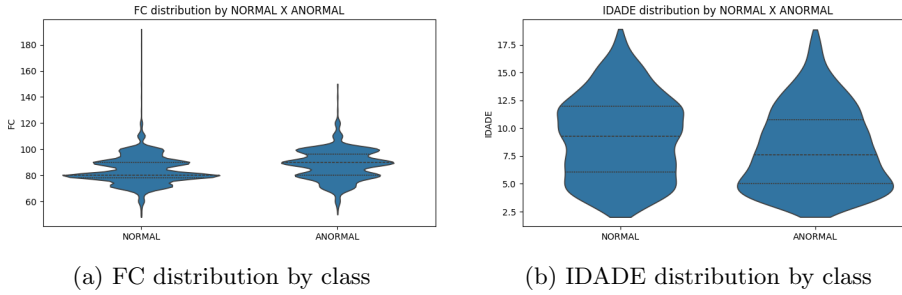


Figure 12: Distribution of a numeric variable split by the target classes (NORMAL vs ANORMAL)

To support interpretability beyond summary statistics, we visualized the standardized mean difference (ANORMAL - NORMAL) across numerical variables

(Figure 11) and inspected class-conditional distributions for the numerical variables showing the clearest group shifts (notably *FC* and *IDADE*) (Figure 12). These plots confirm that, although the effects are statistically consistent, there is substantial overlap between classes, motivating the need for multivariate modeling.

4.1.3 Categorical - Categorical relationships (association and redundancy)

To detect potential redundancy among categorical predictors, we measured pairwise associations between categorical predictors using **Cramér's V**. Cramér's V ranges from 0 (no association) to 1 (perfect association) and is appropriate for nominal variables with multiple categories. For completeness, we also report chi-square *p*-values; however, given the large sample size, we interpret redundancy primarily through effect size (Cramér's V) rather than significance alone.

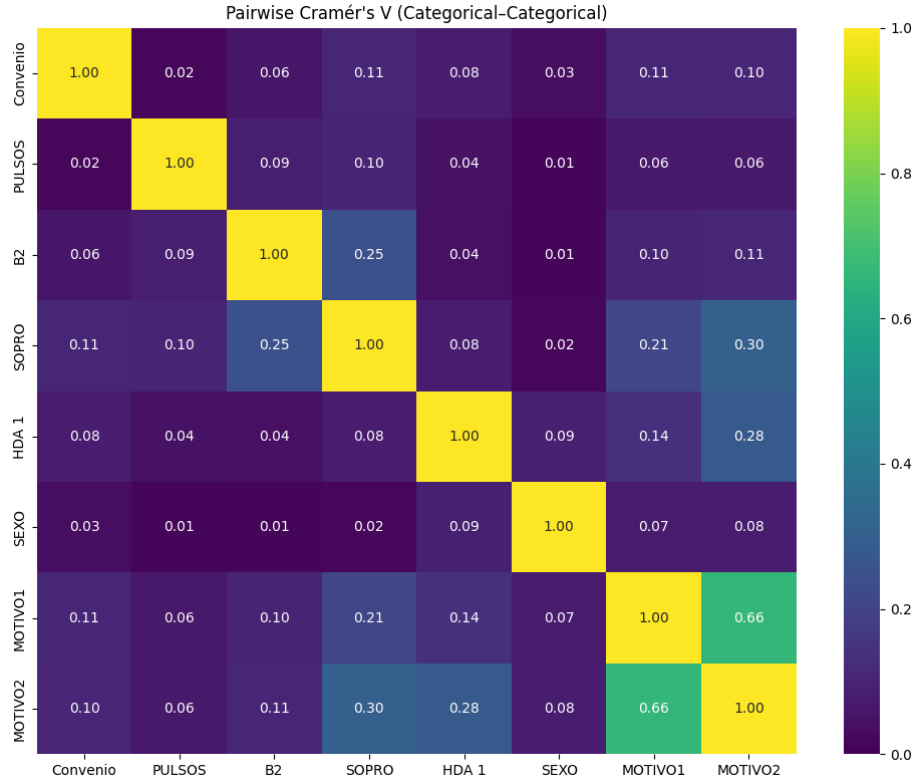


Figure 13: Cramér's V heatmap

Figure 13 shows the pairwise Cramér's V heatmap. Most pairs exhibit **very**

weak association ($V < 0.15$), suggesting that the majority of categorical variables capture largely distinct information. Two moderate clusters stand out. First, MOTIVO1 and MOTIVO2 show a **strong association** ($V \approx 0.66$), indicating substantial overlap between the two recorded reasons for referral and therefore some redundancy. Second, SOPRO displays **moderate** association with MOTIVO2 ($V \approx 0.30$) and MOTIVO1 ($V \approx 0.21$), and B2 is moderately associated with SOPRO ($V \approx 0.25$), which is clinically plausible since several findings and referral reasons are related to suspected cardiac abnormalities. Other associations are weak (e.g., SEXO with all variables, and PULSOS with the remaining clinical fields), suggesting low redundancy in those features.

	VarA	VarB	CramersV	p_value
27	MOTIVO1	MOTIVO2	0.660626	0.000000e+00
21	SOPRO	MOTIVO2	0.299391	3.698308e-260
24	HDA 1	MOTIVO2	0.279805	0.000000e+00
13	B2	SOPRO	0.246054	8.138170e-184
20	SOPRO	MOTIVO1	0.207439	2.085922e-129
23	HDA 1	MOTIVO1	0.136993	3.328940e-125
5	Convenio	MOTIVO1	0.108970	4.309080e-53
17	B2	MOTIVO2	0.106694	1.521469e-61
2	Convenio	SOPRO	0.106347	3.159898e-21
6	Convenio	MOTIVO2	0.102270	2.124033e-129

Figure 14: Top categorical - categorical associations

Table 14 lists the strongest categorical - categorical associations. Based on these results, MOTIVO1 and MOTIVO2 were treated as potentially redundant in subsequent analysis, and multivariate methods were used to handle remaining overlap among clinically related fields.

4.1.4 Categorical - Target relationships

Associations between categorical predictors and the binary outcome *NORMAL* *X* *ANORMAL* were evaluated using contingency tables and the **chi-square test of independence**, complemented by **Cramér’s V** as an effect-size measure. Because the dataset is large, p -values can become extremely small even for weak effects; therefore, variables were primarily ranked by Cramér’s V and interpreted through class-conditional proportions.

	Variable	n_levels	CramersV	chi2_p_value	Top_category_by_ANORMAL_rate	Top_ANORMAL_rate
1	SOPRO	3	0.844019	0.000000e+00	SISTOLICO	0.957766
6	MOTIVO2	16	0.351379	1.090040e-180	5_USO_DE_CISAPRIDA	1.000000
5	MOTIVO1	6	0.263447	2.951991e-106	1_CARDIOPATIA_JA_ESTABELECIDA	0.639024
2	B2	6	0.183959	7.272464e-51	HIPERFONETICA	0.970149
4	Convenio	29	0.155458	3.602429e-23	SUS	1.000000
7	HDA 1	9	0.111137	6.334580e-16	CIANOSE	0.415094
3	PULSOS	6	0.068444	2.511057e-06	DIMINUIDOS	1.000000
0	SEXO	2	0.000000	1.000000e+00	M	0.350857

Figure 15: Table ranking variables by Cramér’s V (categorical - target)

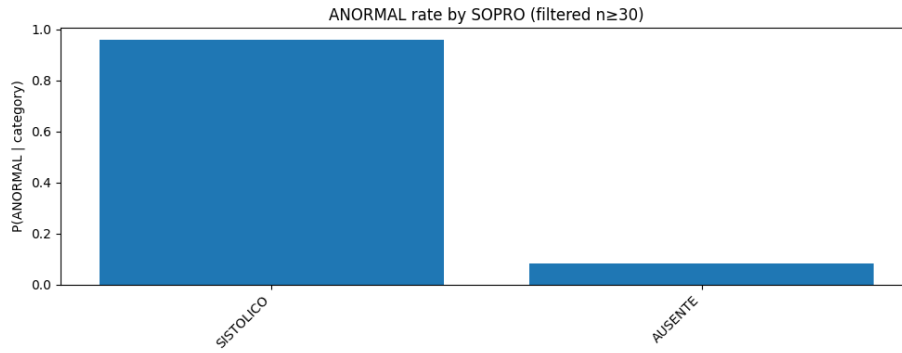


Figure 16: ANORMAL rate by SOPRO filtered

Table 15 summarizes these results. SOPRO exhibits by far the strongest association with the outcome (Cramér’s $V \approx 0.84$), indicating that murmur status is highly informative of the diagnostic category. This is visualized in Figure 16, where *SISTOLICO* is overwhelmingly associated with *ANORMAL* cases (approximately 96%), while *AUSENTE* is predominantly *NORMAL* (approximately 92%).

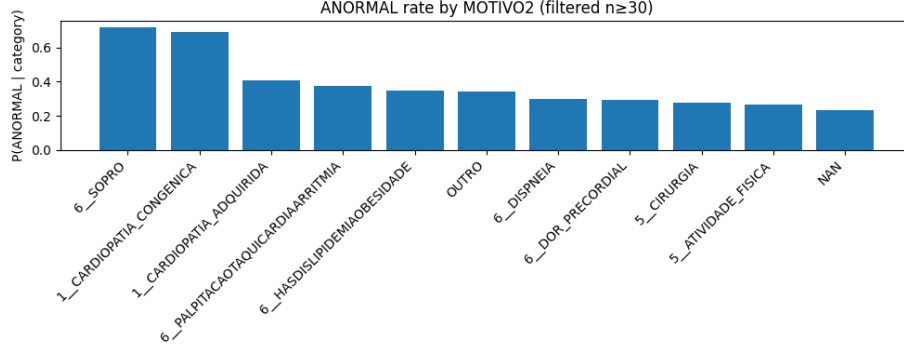


Figure 17: ANORMAL rate by MOTIVO2 filtered

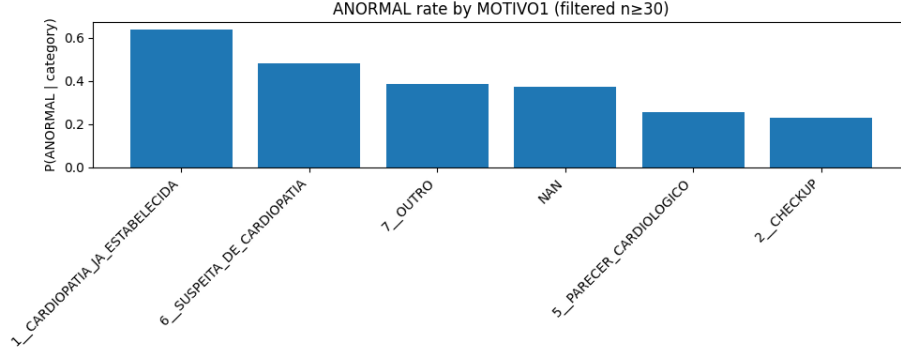


Figure 18: ANORMAL rate by MOTIVO1 filtered

Beyond SOPRO, the reasons for referral (MOTIVO2 and MOTIVO1) show moderate association with the target (Cramér's $V \approx 0.35$ and $V \approx 0.26$, respectively). Figures 17 and 18 display the conditional abnormal rate $P(\text{ANORMAL} | \text{category})$ across their categories, highlighting that some referral reasons correspond to substantially higher abnormal prevalence than routine check-up categories.

Other variables (B2, CONVENIO, HDA1, PULSOS) show smaller effects, suggesting limited stand-alone predictive value but potential usefulness when combined with other features in multivariate models. In contrast, SEXO shows no detectable association (Cramér's $V \approx 0$).

Stability note: To reduce instability from sparse categories, rare levels were consolidated into *OUTROS* and the bar plots were generated after filtering categories with small sample size (e.g., $n \geq 30$). Even with filtering, extreme rates (e.g., 1.0) can occur when a category contains only ANORMAL cases in

the observed sample; these should be interpreted as descriptive evidence rather than deterministic rules.

4.2 Multivariate analysis

While the bivariate analysis evaluates one predictor at a time, clinical discrimination is typically multicausal: predictors can be redundant (e.g., growth variables), interact non-linearly, and only become informative conditionally when considered jointly. Following the task requirements, this section provides a multivariate perspective using (i) mutual information, (ii) multiple-variable regression, and (iii) cluster analysis with a mixed-type dissimilarity appropriate for numerical and categorical variables.

4.2.1 Feature relevance beyond pairwise tests (MI + Random Forest)

To quantify feature relevance beyond pairwise tests and visual inspection, we computed two complementary relevance criteria on the **training split only** (to avoid information leakage), using the same preprocessing intended for modeling: median imputation and standardization for numerical variables, and most-frequent imputation with one-hot encoding for categorical variables.

Split (avoid data leakage): We split the dataset into train and test sets. All relevance scores in this subsection were computed only on the **training set**. The test set is kept untouched for the final model evaluation.

Preprocessing: To make the relevance results consistent with the models later, we used the same preprocessing pipeline:

- Numerical variables: missing values were imputed (median) and then standardized.
- Categorical variables: missing values were imputed (most frequent) and then one-hot encoded.

After one-hot encoding, each category becomes its own binary column (dummy feature).

Mutual Information (MI): Mutual Information measures how much a feature reduces uncertainty about the target. It can capture non-linear relationships. MI is always ≥ 0 and does not tell direction. We computed MI for each encoded feature and then summarized it back to the original variables.

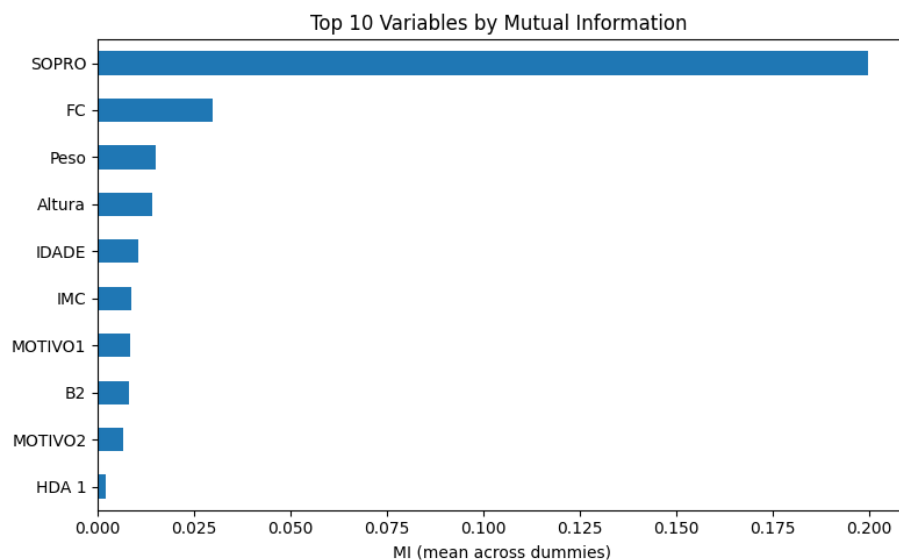


Figure 19: Top-10 variables by Mutual Information (MI) with the target (training split). Categorical variables are one-hot encoded and aggregated to variable level by the mean MI across dummies.

Random Forest importance: Random Forest provides an importance score based on how much each feature helps reduce impurity in the trees. It can capture non-linear effects and interactions. We computed importance for each encoded feature and then summarized it back to the original variables.

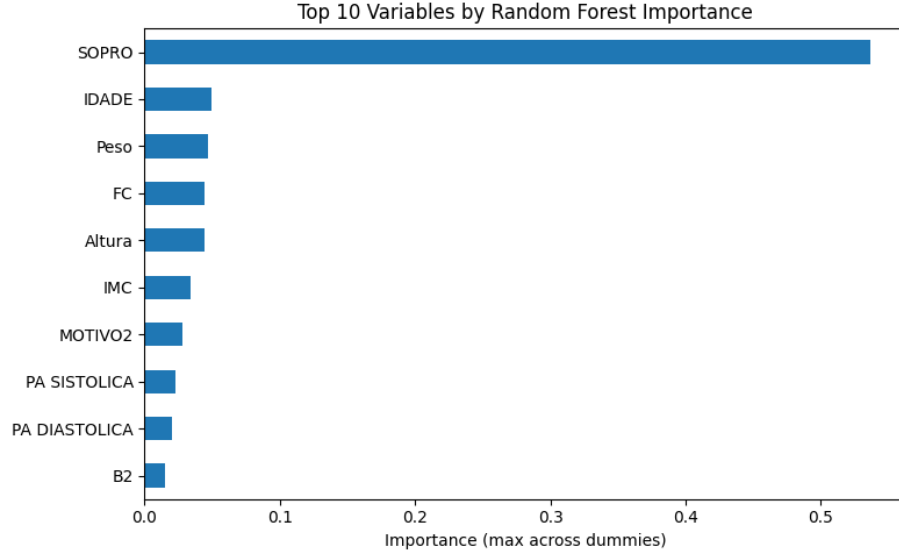


Figure 20: Top-10 variables by Random Forest feature importance (training split). Categorical variables are one-hot encoded and aggregated to variable level by the maximum importance across dummies.

How we go from dummy features to original variables: Because one variable can create many dummy features, we need a simple rule to summarize scores at the variable level:

- For MI, we used the **mean** MI across the dummies of that variable (more “average usefulness”).
- For RF importance, we used the **maximum** importance across the dummies (does the variable contain at least one very strong category?).

This is why the MI and RF rankings can differ slightly for high-cardinality variables (like MOTIVO1 and MOTIVO2).

Why we also use a Combined Score: MI and Random Forest do not measure the same thing. MI measures how strongly a variable is related to the target. Random Forest importance measures how much a variable helps the trees make good splits. Because they can rank variables slightly differently, we build a simple consensus score.

Combined Score (consensus of MI and RF): For each variable, we scale its MI score and its RF importance to the range $[0, 1]$ (min-max scaling). Then we take the average of the two scaled values:

$$\text{Combined Score} = \frac{\text{MI}_{\text{scaled}} + \text{RF}_{\text{scaled}}}{2}.$$

This gives one easy ranking that uses information from both methods and is the main ranking used to define the Top-8 set.

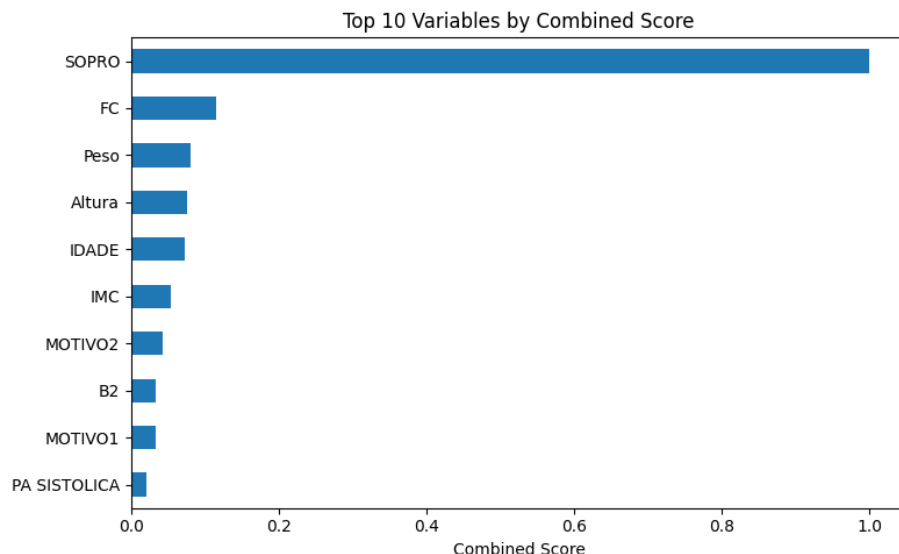


Figure 21: Top-10 variables by Combined Score (training split): min-max scaled MI and RF importance averaged at variable level. Used to define the Top-8 set.

Main result: The Combined Score confirms the same pattern as the previous plots: **SOPRO** is clearly the strongest variable. A second group includes **FC** and growth/context variables such as **IDADE**, **Peso**, and **Altura**. Other variables contribute less.

4.2.2 Multiple-variable regression (joint-effect baseline) and consensus Top-8

The rankings from Section 4.2.1 (MI, Random Forest, and the Combined Score) tell us which variables are generally informative. However, they do not directly show what happens when variables are used *together*. To get a simple multi-variate baseline, we fitted a **logistic regression** model.

Logistic regression setup: We trained logistic regression on the **training set**, using the same preprocessed representation: standardized numerical features + one-hot encoded categorical features. Each coefficient represents how that encoded feature is associated with the probability of being ANORMAL, while the other features are also in the model. Because features are correlated and categorical variables produce many dummies, coefficients should be read as association signals (not causal effects).

What regression adds compared to MI/RF: Logistic regression can highlight variables that matter mostly in combination with others. For example, variables like **B2** or **PULSOS** may not rank very high in marginal MI, but can still contribute when combined with stronger signals.

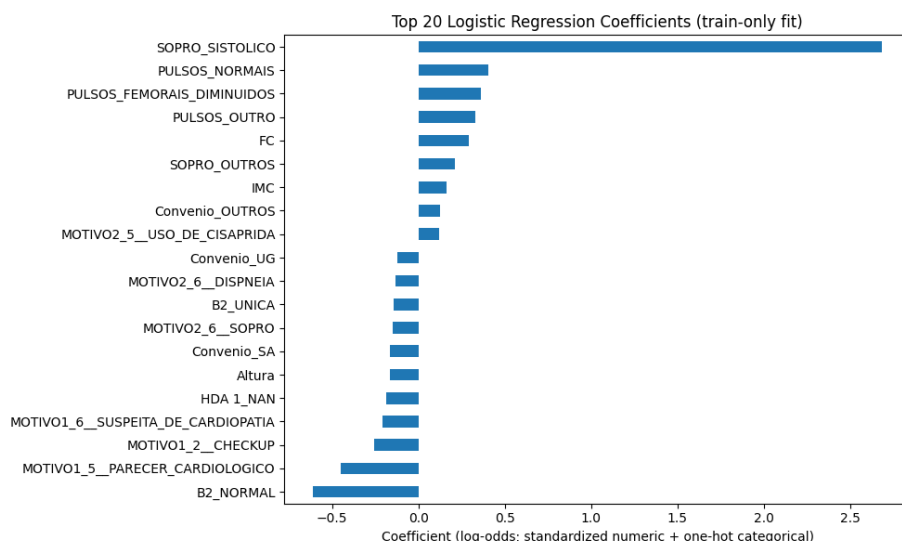


Figure 22: Largest logistic regression coefficients (train only) after preprocessing (standardized numeric + one-hot categorical). Positive coefficients increase the predicted odds of ANORMAL and negative coefficients decrease them, always relative to the reference category omitted in one-hot encoding.

Figure 22 shows the strongest logistic regression weights after one-hot encoding. This view is feature-level (dummy-level), so it highlights which exact categories matter most. The result supports the relevance rankings: **SOPRO_SISTOLICO** has by far the largest positive weight, meaning it strongly increases the predicted odds of ANORMAL compared to the reference murmur category.

Note that with one-hot encoding, each coefficient for a category is interpreted relative to the omitted reference category of the same variable.

Variable-level summary: Because one categorical variable generates many dummy columns, we also summarize regression effects at the variable level by taking the **maximum absolute coefficient** among its dummies. This gives a quick view of which original variables contain at least one strong category.

Top variables by max coefficient across dummies	
SOPRO	2.684996
B2	0.610535
MOTIVO1	0.452335
PULSOS	0.402740
FC	0.289300
HDA 1	0.188864
Altura	0.167804
Convenio	0.163539
IMC	0.163233
MOTIVO2	0.151054
PA SISTOLICA	0.102708
SEXO	0.069591
Peso	0.024172
PA DIASTOLICA	0.014770
IDADE	0.012658

Figure 23: Top-15 variables by max |coefficient|

Consensus idea (MI/RF + regression): At this point we have three complementary views:

- MI: detects dependence with the target (including non-linear patterns).
- Random Forest: captures non-linear splits and interactions.
- Logistic regression: shows which features keep strong effects when fitted together.

We use the Combined Score from Section 4.2.1 as the main ranking, and we use logistic regression as a **sanity check**: variables that appear high in Combined Score and also show strong regression effects are treated as the most reliable candidates. When a variable is high in only one view, we keep it only if it makes clinical sense and does not create redundancy.

	MI rank	RF rank	REG rank	Combined rank
SOPRO	1.0	1.0	1.0	1.0
FC	2.0	4.0	5.0	2.0
B2	8.0	10.0	2.0	3.0
MOTIVO1	7.0	11.0	3.0	4.0
Altura	4.0	5.0	7.0	5.0
Peso	3.0	3.0	13.0	6.0
IMC	6.0	6.0	9.0	7.0
PULSOS	12.0	15.0	4.0	8.0
IDADE	5.0	2.0	15.0	9.0
MOTIVO2	9.0	7.0	10.0	10.0
HDA 1	10.0	12.0	6.0	11.0
Convenio	11.0	14.0	8.0	12.0

Figure 24: Rank-based consensus of feature relevance across Mutual Information (MI), Random Forest (RF), and logistic regression (REG). Combined rank summarizes the three rankings.

Interpretation of the consensus table: Figure 24 compares how the top variables are ranked by three different multivariate views. The strongest agreement appears at the top: **SOPRO** is ranked first by MI, RF, and regression, confirming it as the most consistent signal. **FC** is also consistently near the top, which matches the earlier analyses where heart rate showed the clearest shift between classes. Other variables (such as **B2**, **MOTIV01**, and growth-related measures like **Altura** and **Peso**) rank well but with more variation across methods. This variation is expected because each method captures a different type of information: MI focuses on dependence with the target, RF captures non-linear splits/interactions, and regression highlights strong effects when features are used together.

During the rest of the Multivariate Analyses, we will have this consensus table in mind together with the Combined Score ranking from Section 4.2.1 and the redundancy information from the bivariate analysis (e.g., overlap between **Peso/Altura/IMC**) to decide which feature sets to test. The goal is to choose a small set of variables that keeps most of the predictive signal while avoiding unnecessary overlap.

4.2.3 Cluster analysis (patient similarity structure)

The previous subsections focus on feature relevance for supervised prediction. Here, we use clustering as an exploratory check: we want to see if patients form well-separated groups in the full feature space (numeric + categorical). Because the dataset contains mixed variable types, we use a mixed-type distance.

Distance for mixed data (Gower): We compute a Gower-style dissimilarity between patients by combining: (i) normalized absolute differences for numeric variables, and (ii) mismatch distances for categorical variables (0 if equal, 1 if different). The final distance is the average of the numeric and categorical components.

Train-only clustering to avoid leakage: Clustering is performed using the training split only. We build the clustering matrix from the original variables (excluding the target), and we impute missing values using simple train-only rules: median for numeric variables and mode for categorical variables. This keeps the clustering step consistent with the idea of not using test data during exploratory structure discovery.

Clustering algorithm and number of clusters: We apply hierarchical (agglomerative) clustering on the precomputed Gower distance matrix. To choose the number of clusters, we compute the silhouette score for $k = 2$ to $k = 6$.

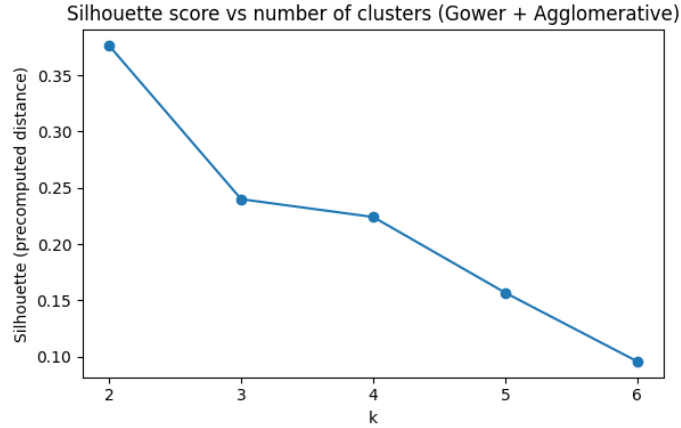


Figure 25: Silhouette score for hierarchical clustering using Gower distance (train only).

In our results, the silhouette score is highest for $k = 2$ (approximately 0.38) and decreases steadily for larger k , suggesting that the data does not naturally separate into multiple stable clusters under this distance.

Cluster sizes and clinical meaning: Using the best silhouette solution ($k = 2$), the clustering produces one very large cluster and a second cluster with only a few patients (outlier-like behavior).

	n	abnormal_rate
Cluster		
0	5784	0.350622
1	2	0.500000

Figure 26: Cluster sizes (n) for the selected hierarchical clustering solution (train only).

We also checked whether clusters differ in target prevalence. The large cluster has an ANORMAL rate around 0.35, while the tiny cluster has an ANORMAL rate of 0.50, but this value is not meaningful due to the extremely small size of that cluster.

Cluster profiles: To understand what the clustering is capturing, we computed simple cluster profiles: mean values for numeric variables and the most frequent category (mode) for categorical variables. The tiny cluster shows unusual combinations of values and categories, consistent with an outlier split, rather than a clinically meaningful subgroup.

	Peso	Altura	IMC	IDADE	PA SISTOLICA	PA DIASTOLICA	FC
Cluster							
0	33.001798	130.39834	18.409751	8.845325	100.97666	62.151971	85.527361
1	14.000000	94.00000	16.500000	3.195000	105.00000	70.000000	115.000000

Figure 27: Numeric cluster profile for the selected $k = 2$ solution (train only): mean values of the main numerical variables per cluster.

	Convenio	PULSOS	B2	SOPRO	HDA 1	SEXO	MOTIVO1	MOTIVO2
Cluster								
0	NAN	NORMAIS	NORMAL	AUSENTE	ASSINTOMATICO	M	5_PARECER_CARDIOLOGICO	NAN
1	MEDIAL	AMPLOS	NORMAL	OUTROS	DISPNEIA	F	1_CARDIOPATIA_JA_ESTABELECIDA	1_CARDIOPATIA_CONGENICA

Figure 28: Categorical cluster profile for the selected $k = 2$ solution (train only): most frequent category (mode) of each categorical variable per cluster.

Interpretation: Overall, clustering does not provide evidence of stable, clinically meaningful patient groups in this dataset under Gower distance and hierarchical clustering. The main structure is one dominant cluster plus a very small outlier cluster, and cluster-level abnormal prevalence does not show a clear separation. Therefore, clustering is used here as a multivariate structure check, not as the main driver of feature selection.

4.2.4 Multivariate summary and modeling hypotheses

This multivariate analysis combined three supervised views (MI, Random Forest, logistic regression) and one unsupervised view (clustering). Our goal was not to pick a single “best” list immediately, but to build clear evidence about which variables repeatedly carry signal and which ones are likely redundant.

Main signals supported by multiple methods: Across MI, RF, the Combined Score (Section 4.2.1), and regression coefficients (Section 4.2.2), SOPRO is consistently the strongest predictor. FC also appears near the top in several views. Other variables such as B2, PULSOS, MOTIVO1/MOTIVO2, and growth-related measures (IDADE, Peso, Altura, IMC) show useful but more variable contributions depending on the method.

Redundancy and practical constraints: Some variables overlap by definition or by recorded meaning. For example, IMC is derived from **Peso** and **Altura**, and MOTIVO1 and MOTIVO2 are strongly associated. Because of this, the feature sets tested in modeling should avoid unnecessary overlap where possible, and should include only what is needed to keep performance while improving simplicity.

What clustering adds (and what it does not): Clustering using Gower distance did not reveal clear, stable patient subgroups. The best silhouette

solution splits the data into one dominant cluster and a very small outlier-like cluster. The cluster profiles (Figures 27 - 28) mainly support the idea that clustering is capturing rare patterns rather than clinically meaningful groups. Therefore, clustering is used as a structure check, not as the main driver of feature selection.

Feature sets to be evaluated in modeling: Instead of fixing a single final list purely from one ranking, we define three feature configurations for Section 4.3. These configurations are guided by: (i) the Combined Score ranking from Section 4.2.1 (Figure 21), (ii) the MI/RF/REG rank-consensus table from Section 4.2.2 (Figure 24), and (iii) redundancy findings from the bivariate analysis (e.g., overlap between *Peso/Altura/IMC* and the association between *MOTIV01* and *MOTIV02*).

Concretely, we test:

- **Full feature set (Collection I):** all predictors after preprocessing (baseline).
- **Optimized Top-8 (Collection II):**

$\{\text{SOPRO}, \text{FC}, \text{IDADE}, \text{Peso}, \text{Altura}, \text{B2}, \text{MOTIV01}, \text{MOTIV02}\}.$

This set was chosen because these variables appear repeatedly near the top across the Combined Score and the rank-consensus views, and it keeps the model simpler while preserving the main signal. We also avoid including IMC together with *Peso* and *Altura* to reduce redundancy.

- **No SOPRO (Collection III):** the full feature space excluding *SOPRO*, used to quantify how much performance depends on murmur information.

In addition, small Top-8 variations were considered during development. Since modeling results were very similar across these compact variants, we kept the Top-8 above to ensure consistent comparisons across models.

Hypotheses before modeling: Based on the multivariate results, we expect:

- **H1:** The compact feature set will achieve performance close to the full feature set, because the predictive signal is concentrated in a small number of variables.
- **H2:** Removing *SOPRO* will cause a noticeable performance drop, especially in recall for the ANORMAL class, since murmur information dominates all relevance rankings.
- **H3:** Variables like *FC* and referral motives (*MOTIV01/MOTIV02*) will help refine predictions when combined with other features, even if they are weaker than *SOPRO*.

This closes the multivariate analysis and motivates the modeling experiments that follow.

4.3 Predictive Models

To evaluate the predictive power of various machine learning algorithms in detecting pediatric cardiac conditions, we implemented five distinct models: **Logistic Regression**, **Random Forest**, **XGBoost**, **Support Vector Machine (SVM)**, and a **Multi-Layer Perceptron (MLP)**. The performance of these models was assessed using a comprehensive suite of metrics: **Accuracy** for overall correctness, **Precision** to measure diagnostic certainty, **Recall** to evaluate the ability to identify all positive cases, and the **F1-Score** and **ROC-AUC** to determine the balance between sensitivity and specificity.

The analysis was conducted across three distinct data configurations to observe how feature selection and the removal of clinical indicators, specifically the presence of a murmur, impact diagnostic reliability.

4.3.1 Model Collection I: Full Feature Set

Random Forest stands as the most accurate and precise model, making it the most reliable choice for confirming a diagnosis while minimizing the emotional and financial cost of false alarms. However, from a clinical safety standpoint, the MLP is the superior screening tool because its higher recall ensures fewer sick patients are overlooked during initial triage. While Random Forest provides the best overall predictive power with the highest ROC-AUC, the MLP’s ability to catch more positive cases is critical in a medical setting where a missed diagnosis is far more dangerous than a false positive. Because every model still fails to identify approximately 18% to 20% of cases, these results should be used to help physicians prioritize high-risk patients rather than acting as a final diagnostic authority.

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.918452	0.949309	0.811024	0.874735	0.918776
1	Random Forest	0.919834	0.955814	0.809055	0.876333	0.930143
2	XGBoost	0.914305	0.936364	0.811024	0.869198	0.924721
3	SVM	0.912923	0.934091	0.809055	0.867089	0.916971
4	MLn	0.906704	0.906318	0.818898	0.860393	0.917048

Figure 29: Model performance (test set) for Collection I: Full feature set.

4.3.2 Model Collection II: Optimized Top 8 Features

Logistic Regression emerges as the most accurate and precise model when using the top 8 selected features, making it the most dependable choice for confirming a diagnosis and minimizing the psychological and clinical burden of false positives. However, from a safety-first perspective, the SVM is the most effective screening tool in this configuration because its higher recall ensures that fewer sick patients are overlooked during the diagnostic process. While Random Forest and Logistic Regression provide excellent diagnostic certainty, the SVM’s superior ability to detect actual cases is critical in a medical context where the cost of a missed diagnosis outweighs the inconvenience of a false alarm. Because

all models still fail to identify approximately 18% to 20% of positive cases, they should be used as decision-support systems to help doctors prioritize high-risk patients rather than as a final diagnostic authority.

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.917761	0.947126	0.811024	0.873807	0.925274
1	Random Forest	0.915688	0.944700	0.807087	0.870488	0.917646
2	XGBoost	0.914997	0.940503	0.809055	0.869841	0.915569
3	SVM	0.917070	0.938914	0.816929	0.873684	0.913168
4	Mln	0.908086	0.921348	0.807087	0.860441	0.917151

Figure 30: Model performance (test set) for Collection II: Top-8 feature set.

4.3.3 Model Collection III: Impact of Excluding Clinical Murmur Findings

Logistic Regression and Random Forest are the most accurate models in this set, while SVM provides the highest precision, making it the most reliable choice for avoiding the stress of false alarms. However, from a clinical safety perspective, the MLP remains the most effective screening tool because its higher recall ensures that fewer sick patients are overlooked compared to the other models. It is critical to note that performance has dropped significantly without the "murmur" feature, with all models now failing to identify between 41% and 58% of positive cases. Because of this high miss rate, these models are not yet suitable for primary diagnosis and instead highlight the essential value of clinical findings like murmurs in successfully detecting cardiac conditions.

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.738079	0.696049	0.450787	0.547192	0.761587
1	Random Forest	0.747754	0.698061	0.496063	0.579977	0.769963
2	XGBoost	0.713891	0.615196	0.494094	0.548035	0.740734
3	SVM	0.736006	0.710000	0.419291	0.527228	0.750130
4	Mln	0.705598	0.579457	0.588583	0.583984	0.731115

Figure 31: Model performance (test set) for Collection III: Full set without SOPRO.

4.4 Comparison

The following analysis compares the three experimental configurations to determine the impact of feature engineering and the clinical significance of specific diagnostic markers on model performance, focusing on the trade-off between sensitivity and diagnostic certainty.

4.4.1 Feature Optimization: Full Set vs. Top 8 Features

The transition from the Full Feature Set to the Top 8 Optimized Features resulted in a marginal change in overall metrics, yet the models remained remarkably robust. The **ROC-AUC** remained high (above 0.91), indicating that the

models’ ability to distinguish between healthy and sick patients was not significantly compromised by reducing the feature space. **Precision** remains high, given that Random Forest achieves the highest precision in the full set (0.956), while Logistic Regression has the highest precision in the Top-8 set (0.947) and the best **Recall** is similar across the two settings: MLP reaches 0.819 in the full set, while SVM reaches 0.817 in the Top-8 set. This suggests that the ”signal” for cardiac conditions is concentrated in these top features; however, even with an optimized set, the models consistently leave a 18%–20% ”blind spot” in detection.

4.4.2 The Critical Role of Clinical Findings: The ”Murmur” Effect

The most significant finding occurred upon the removal of the ”murmur” feature. In Model Collection III, the **ROC-AUC** drops from about 0.93 (full set) to about 0.77 (no SOPRO), representing a massive loss in the models’ discriminative power. More critically, **Recall** drops to roughly 42%–59%, meaning the models failed to identify nearly half of all positive cases. While **Precision** also decreased, the catastrophic drop in Recall is the most concerning clinical outcome, as it indicates that without murmur data, the models become unsafe for screening. This highlights that while ML can process complex data, it cannot yet overcome the loss of a primary clinical indicator that carries the highest weight in the Recall calculation.

4.4.3 Synthesis and Clinical Recommendation

The comparison across all three collections demonstrates a clear hierarchy of utility driven by the Precision-Recall balance. For a screening tool, **Recall** is the priority; MLP leads recall in the full set and in the no-SOPRO setting, while SVM leads recall in the Top-8 setting., making these models the best choices for ensuring patient safety by minimizing false negatives. Conversely, **Precision** is the priority for diagnosis to avoid unnecessary invasive follow-ups; here, Logistic Regression and Random Forest excelled, maintaining high certainty even with fewer features.

Ultimately, the **Top 8 Features (including Murmur)** configuration provides the best balance of efficiency and performance. We recommend a dual-model approach: using the **MLP** for initial high-sensitivity screening to ”catch” as many cases as possible, followed by **Logistic Regression** to confirm those cases with high precision. However, because the ROC-AUC and Recall both suffer significantly without clinical findings like murmurs, these models must be viewed as an enhancement to - not a replacement for - physical examinations.

5 Discussion and Main Conclusions

5.1 Discussion

This project explored the prediction of pediatric cardiac pathology (NORMAL vs ANORMAL) using a cleaned clinical dataset with demographic, growth, vital-sign, and clinical finding variables. The workflow followed a standard data science pipeline: (i) identify and fix raw-data issues (outliers, invalid ranges, label noise), (ii) describe the cleaned dataset, (iii) perform bivariate and multivariate analyses to understand relevance and redundancy, and (iv) evaluate multiple predictive models under different feature configurations.

What the analyses consistently show: Across bivariate results, multivariate relevance rankings (MI, Random Forest, regression), and the modeling experiments, the variable `SOPRO` (murmur type) is the dominant predictor. This is clinically plausible: a murmur is a direct finding linked to cardiac abnormality and, in this dataset, the `SISTOLICO` murmur category is strongly associated with the `ANORMAL` class. Heart rate (`FC`) is the most informative numerical feature, but numerical variables alone show substantial overlap between classes, indicating that no single vital sign provides strong discrimination in isolation.

Feature reduction and robustness: Reducing the feature space from the full set to the Top-8 set produced only minor changes in performance. ROC-AUC remained high (above 0.91), and the best recall stayed near ~ 0.82 . This suggests that a large part of the predictive signal is concentrated in a small set of clinically meaningful variables, and that a compact feature set can preserve most of the model’s discriminative ability while improving simplicity and interpretability.

The “murmur” effect: The most important practical result is the sharp performance degradation when `SOPRO` is removed. In the no-`SOPRO` configuration, ROC-AUC drops to around 0.77 and recall falls to roughly 42%–59%, meaning that a large fraction of sick patients would be missed. This confirms that the models depend heavily on murmur information and reinforces the clinical reality that physical examination findings remain central for reliable screening.

Clinical interpretation of model trade-offs: The models show a consistent trade-off between diagnostic certainty (precision) and safety (recall). In a screening context, recall is the priority because false negatives are costly. In our experiments, recall leaders vary by configuration (e.g., MLP in the full set and no-`SOPRO` setting; SVM in the Top-8 setting). In a confirmation context, high precision is more desirable to reduce unnecessary follow-ups, where Logistic Regression and Random Forest perform strongly. This supports the idea that model choice should reflect the clinical stage (screening vs confirmation) rather than selecting a single algorithm as universally “best”.

Limitations: Several limitations affect interpretation:

- **Dataset scope:** the data comes from a specific clinical context and may not generalize to other hospitals or populations without external validation.
- **Class overlap and residual uncertainty:** even with strong predictors, the best models still leave a noticeable miss-rate (roughly an 18%–20% “blind spot” in the full and Top-8 settings), indicating real overlap in observed measurements and/or label noise.
- **High-cardinality categorical variables:** variables like `MOTIV01`/`MOTIV02` contain heterogeneous categories; their contribution may depend on how rare categories were grouped and on how well those categories represent stable clinical concepts.
- **Interpretation limits:** feature relevance and regression coefficients capture association, not causality. Coefficients for one-hot categories are also relative to an omitted reference category.

5.2 Main Conclusions

- The dataset required extensive preprocessing (range filtering, label standardization, removal of redundant/high-missingness fields), and the cleaned dataset supports meaningful analysis and modeling.
- Multivariate evidence consistently identifies `SOPRO` as the strongest predictor of pediatric cardiac abnormality, with `FC` and growth/context variables providing additional (but weaker) support.
- A compact Top-8 feature set preserves performance close to the full feature set (ROC-AUC above 0.91), supporting the idea that the predictive signal is concentrated and that simpler models can be competitive.
- Removing `SOPRO` causes a major performance drop (ROC-AUC around 0.77 and recall dropping to roughly 42%–59%), showing that murmur information is essential for safe screening in this dataset.
- The best model depends on clinical use: high-recall models are preferable for screening, while high-precision models are preferable for confirmation. In all cases, the models should be used as **decision support** and not as a replacement for physical examination.